

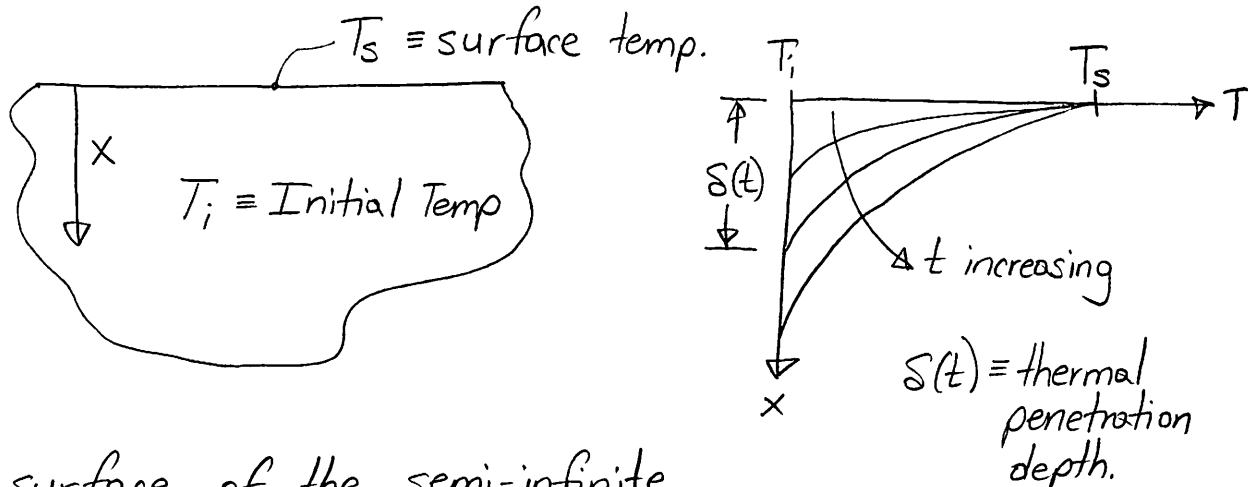
Transient Conduction

So far, we have been mainly dealing with steady-state conduction problems. What if transient effects are important and $Bi > 0.1$?

$$\frac{\partial^2 T}{\partial x^2} + \underbrace{\frac{\partial^2 T}{\partial y^2}}_{0 \text{ to } 10} + \underbrace{\frac{\partial^2 T}{\partial z^2}}_{0 \text{ to } 10} + \underbrace{\frac{Q'''}{k}}_{0 (Q'''=0)} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \Rightarrow \text{Second order PDE. Need 2 BC's \& 1 IC}$$

Semi-Infinite Body



The surface of the semi-infinite body suddenly experiences a finite temperature (T_s), how does the body temperature $T(x)$ change with time (t)?

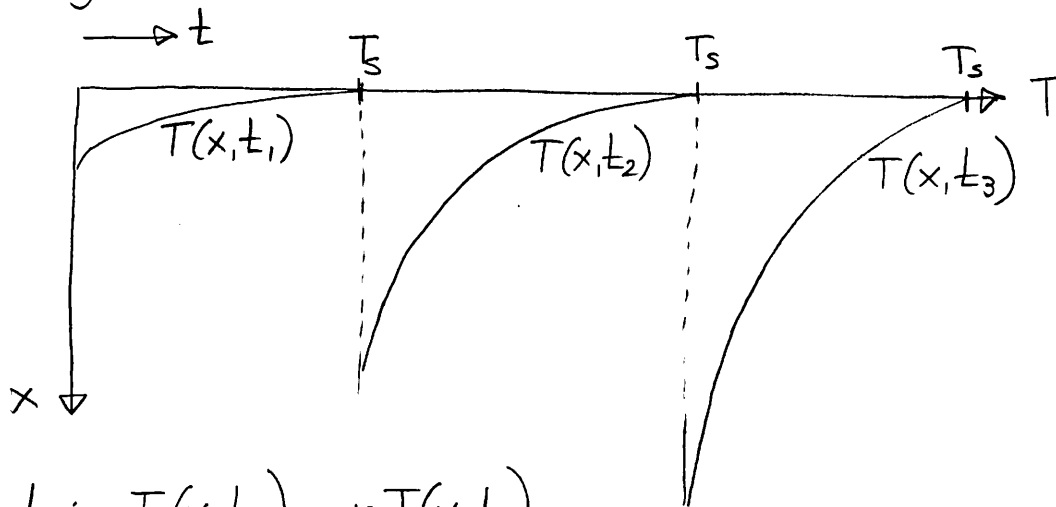
Let's non dimensionalise our temperature & solve our PDE:

$$\Theta = \frac{T - T_s}{T_i - T_s} \quad \left\{ \begin{array}{l} 1) \ x=0, \ \Theta=0 \ (T(x=0)=T_s) \\ 2) \ x \rightarrow \infty, \ \Theta=1 \ (T(x \rightarrow \infty)=T_i) \\ 3) \ t=0, \ \Theta=1 \ (T(x, t=0)=T_i) \end{array} \right.$$

$$\frac{\partial^2 \Theta}{\partial x^2} = \frac{1}{\alpha} \frac{\partial \Theta}{\partial t}$$

↳ Note, separation of variables won't work due to infinite med. (60)

The best way to solve this equation is to look for a scaling parameter or a similarity variable.



Note: $T(x, t_2) = nT(x, t_1)$
 $T(x, t_3) = nT(x, t_2) = mT(x, t_1)$

All of the temperature profiles look self similar in nature, just scaled up or down. This tells us there is a fundamental underlying similarity variable:

Assume: $\eta = \frac{x}{f(t)}$, so $\Theta(\eta)$, $f(t) \equiv$ function of time

Transforming our equation:

$$\frac{\partial \Theta}{\partial x} = \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} = \frac{1}{f} \frac{\partial \Theta}{\partial \eta}$$

$$\frac{\partial^2 \Theta}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial \Theta}{\partial x} \right) = \frac{1}{f} \frac{\partial}{\partial x} \left(\frac{\partial \Theta}{\partial \eta} \right) = \frac{1}{f} \frac{\partial}{\partial \eta} \left(\frac{\partial \Theta}{\partial \eta} \right) \frac{\partial \eta}{\partial x} = \frac{1}{f^2} \frac{\partial^2 \Theta}{\partial \eta^2}$$

$$\frac{\partial \Theta}{\partial t} = \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial t} = -\frac{\partial \Theta}{\partial \eta} \cdot x \cdot \frac{f'}{f^2} = -\eta \frac{\partial \Theta}{\partial \eta} \frac{f'}{f}$$

Back substituting into our POE

$$\frac{1}{f^2} \frac{\partial^2 \theta}{\partial \eta^2} + \frac{f'}{f\alpha} \eta \frac{\partial \theta}{\partial \eta} = 0$$

$$\frac{\partial^2 \theta}{\partial \eta^2} + \underbrace{\left(\frac{f \cdot f'}{\alpha} \right)}_{\text{Let this equal to a constant} = 2} \eta \frac{\partial \theta}{\partial \eta} = 0$$

Let this equal to a constant = 2 (could choose any number)

$$\frac{ff'}{\alpha} = 2$$

$$f \frac{df}{dt} = 2\alpha \Rightarrow \int f df = \int 2\alpha dt$$

$$\frac{f^2}{2} = 2\alpha t$$

$$\boxed{f = 2\sqrt{\alpha t}} \Rightarrow \boxed{\eta = \frac{x}{2\sqrt{\alpha t}}}$$

↳ Note, this may seem arbitrary but it ends up working out & providing a solution.

So our equation becomes:

$$\frac{\partial^2 \theta}{\partial \eta^2} + 2\eta \frac{\partial \theta}{\partial \eta} = 0 \Rightarrow \text{ODE only! Nice!}$$

B.C.'s: $\left. \begin{array}{l} \eta = 0, \theta = 0 \\ \eta \rightarrow \infty, \theta = 1 \end{array} \right\} \text{Turned our 2BC's \& IC into 2BC's only.}$

Rewriting our ODE:

$$\frac{\theta''}{\theta'} = 2\eta$$

$$\frac{2}{\partial \eta} (\ln \theta') = \frac{\theta''}{\theta'}$$

$$\frac{2}{\partial \eta} (\ln \theta') = -2\eta \quad (\text{Integrate})$$

$$\ln(\theta') = -\frac{2\eta^2}{2} + C = -\eta^2 + C, \quad (\text{take exponent on both sides})$$

$$\theta' = C_2 e^{-\eta^2}$$

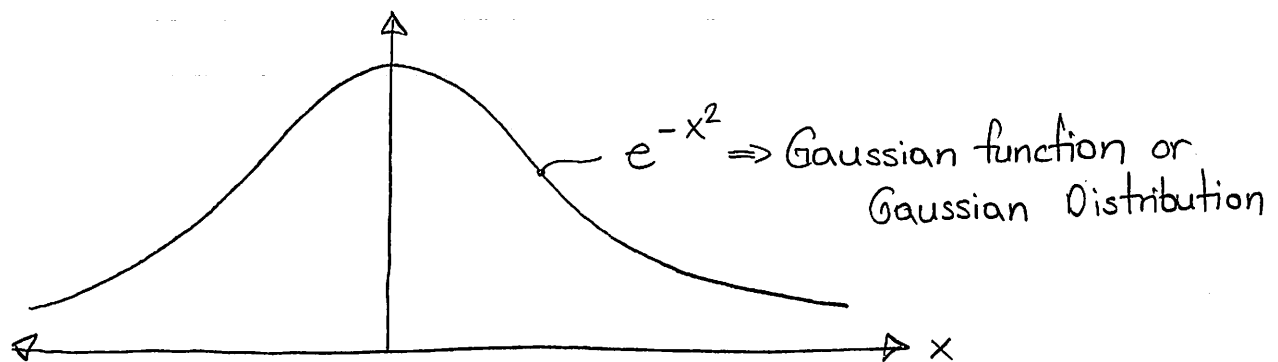
Integrating one more time:

$$\Theta = C_2 \int_0^{\eta} e^{-\eta^2} d\eta + C_3$$

Applying our B.C.'s

$$\Theta(\eta=0) = 0 \Rightarrow \boxed{C_3 = 0}$$

$$\Theta(\eta \rightarrow \infty) = 1 \Rightarrow 1 = C_2 \underbrace{\int_0^{\infty} e^{-\eta^2} d\eta}_{\text{Let's chat about this a bit!}}$$



The function describes "error" or difference between a measurement and its unbiased estimator (or mean).

$\int_0^x e^{-x^2} dx$ cannot be calculated analytically, but it can be shown analytically that:

$$\boxed{\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}} \Rightarrow \boxed{\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}} \Rightarrow \text{Symmetric function about } x=0$$

So now we can define the following:

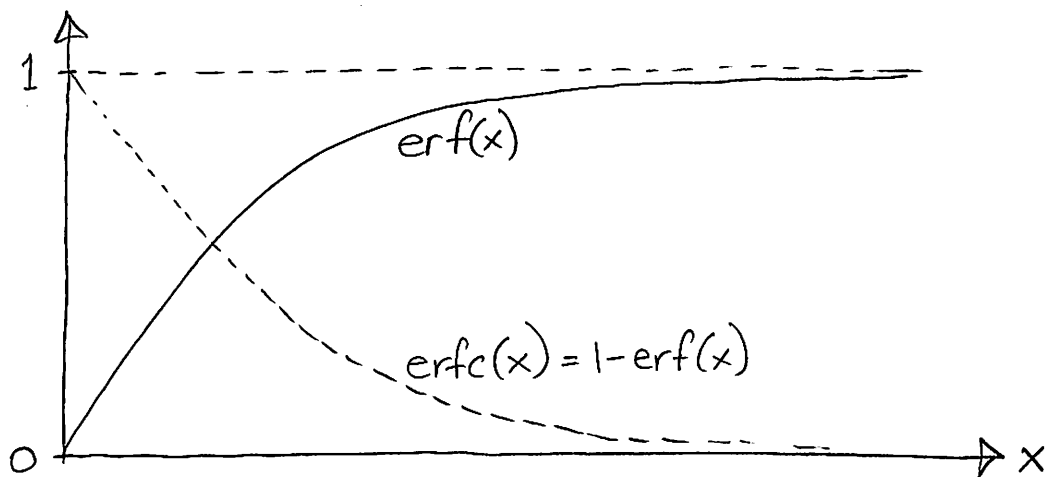
We define the "error" function as:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Error Function

$$\text{erfc}(x) = 1 - \text{erf}(x)$$

Complimentary Error Function



So now we can go back and solve. We had the following:

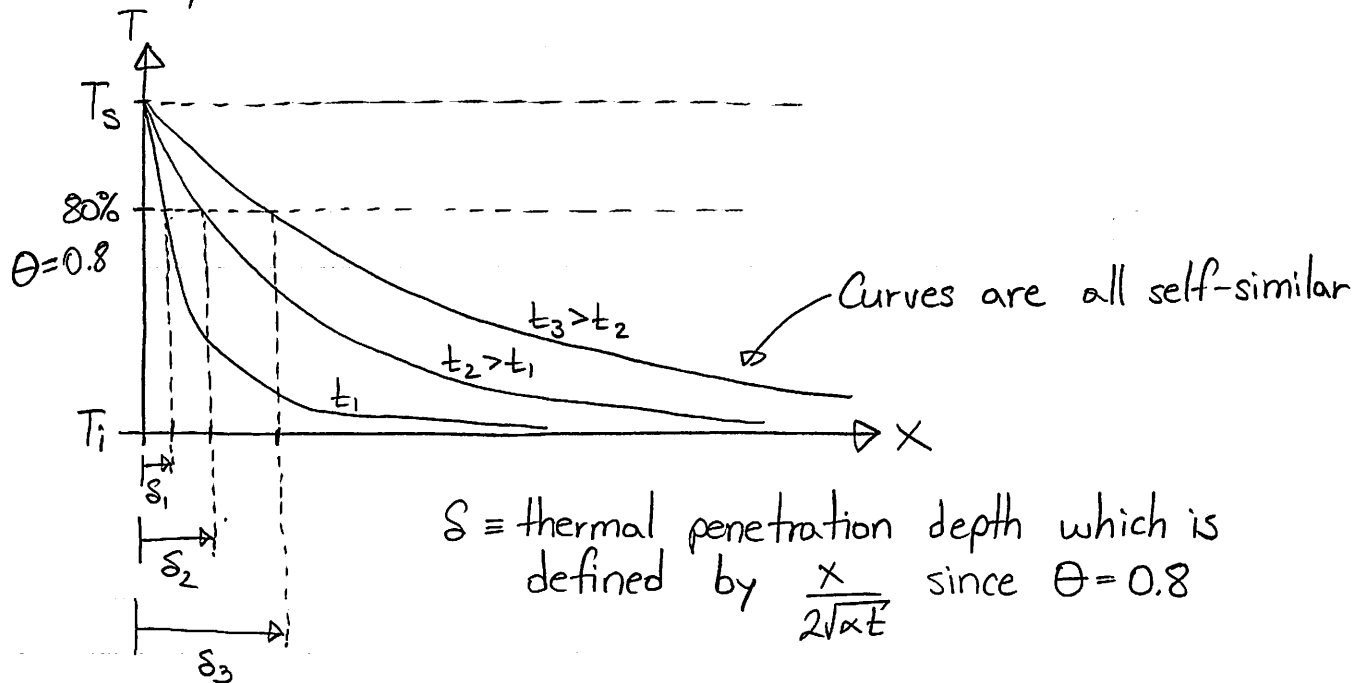
$$1 = C_2 \int_0^{\infty} e^{-\eta^2} d\eta$$

$$\frac{\sqrt{\pi}}{2} \Rightarrow \boxed{C_2 = \frac{2}{\sqrt{\pi}}}$$

$$\therefore \boxed{\Theta = \frac{T - T_s}{T_i - T_s} = \frac{2}{\sqrt{\pi}} \int_0^{\eta} e^{-\eta^2} d\eta = \text{erf}(\eta) = \text{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right)}$$

↳ Tabulated results for $\text{erf}(\eta)$ on pg. 1015 of Textbook
Appendix B, Table B.2

If we plot our result:



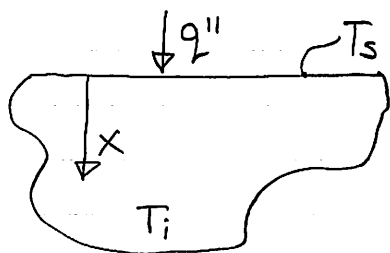
In our equation, we had:

$$\Theta = \frac{T - T_s}{T_i - T_s} = 0.8 = \operatorname{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right) = \operatorname{erf}\left(\frac{\delta}{2\sqrt{\alpha t}}\right)$$

$$\frac{\delta}{2\sqrt{\alpha t}} = \text{constant} \Rightarrow \delta \sim 2\sqrt{\alpha t}$$

So we can say that the heat penetration depth proceeds with $\sqrt{\alpha t}$. Usually we say δ is found at $\Theta = 0.95$ or 0.99 .

How do we determine heat flux?



$$q''|_{x=0} = -k \left. \frac{\partial T}{\partial x} \right|_{x=0}$$

$$\Theta = \frac{T - T_s}{T_i - T_s} ; d\Theta = \frac{1}{T_i - T_s} dT$$